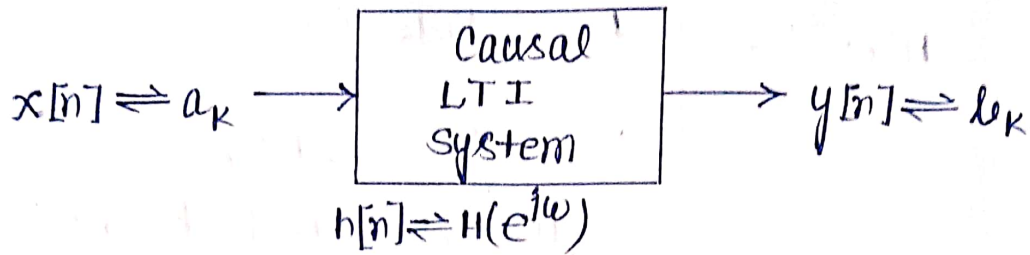


S&S QUIZ-6 SOLUTION

Sol(1)÷



If $x[n]$ is periodic with ' N_0 ' then—

$$x[n] = \sum_{k=\langle N_0 \rangle} a_k \cdot e^{jk\omega_0 n} \quad \text{--- (1)}$$

therefore, $y[n] = x[n] * h[n]$

$$\begin{aligned} &= \sum_{l=-\infty}^{\infty} x[n-l] h[l] \\ &= \sum_{l=-\infty}^{\infty} h[l] \cdot \sum_{k=\langle N_0 \rangle} a_k \cdot e^{jk\omega_0(n-l)} \\ &= \sum_{k=\langle N_0 \rangle} a_k \cdot e^{jk\omega_0 n} \cdot \sum_{l=-\infty}^{\infty} h[l] \cdot e^{-jk\omega_0 l} \\ &= \sum_{k=\langle N_0 \rangle} a_k \cdot e^{jk\omega_0 n} \cdot H(e^{jk\omega_0}) \\ &= \sum_{k=\langle N_0 \rangle} a_k \cdot H(e^{jk\omega_0}) \cdot e^{jk\omega_0 n} \quad \text{--- (2)} \end{aligned}$$

→ (1 Marks)

By eqⁿ(2), Fourier series coefficient of $y[n]$,

$$b_k = a_k \cdot H(e^{jk\omega_0}) \quad \text{--- (3)}$$

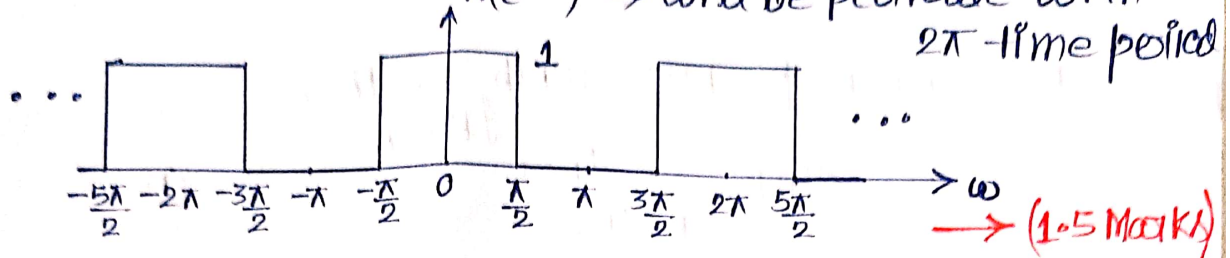
$$\& \quad y[n] = \sum_{k=\langle N_0 \rangle} b_k \cdot e^{jk\omega_0 n} \quad \text{--- (4)}$$

NOTE÷ Input $x[n]$ & output $y[n]$ will be periodic with same time period.

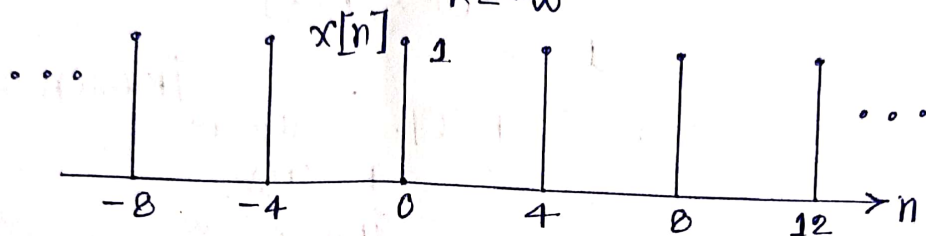
Given frequency response of LTI system,

$$H(e^{j\omega}) = \begin{cases} 1 & |\omega| \leq \pi/2 \\ 0 & \pi/2 < |\omega| < \pi \end{cases}$$

$H(e^{j\omega}) \Rightarrow$ will be periodic with 2π -time period



Given input signal, $x[n] = \sum_{k=-\infty}^{\infty} \delta[n-4k] \iff a_k$



$$\therefore a_k = \frac{1}{N} \sum_{n \in \langle N \rangle} x[n] \cdot e^{-jk\omega_0 n}$$

Here, $N=4$

$$\omega_0 = \frac{2\pi}{N} = \frac{\pi}{2}$$

$$= \frac{1}{4} \sum_{n \in \langle 4 \rangle} x[n] \cdot e^{-jk \frac{\pi}{2} n}$$

$$= \frac{1}{4} \sum_{n=0}^4 \delta[n] \cdot e^{-jk \frac{\pi}{2} n}$$

$$= \frac{1}{4} \quad \forall k \quad \rightarrow (1 \text{ Mark})$$

$$\therefore b_k = a_k \cdot H(e^{jk\omega_0}) \quad \& \quad b_k = b_{k+mN}$$

$$= \frac{1}{4} \cdot H(e^{jk \frac{\pi}{2}})$$

$$b_0 = \frac{1}{4}, b_1 = \frac{1}{4}, b_2 = 0, b_3 = \frac{1}{4}, b_4 = \frac{1}{4}, b_5 = \frac{1}{4}, \dots$$

$$\therefore b_0 = b_4 = b_8 = b_{12} = \dots = \frac{1}{4}$$

$$\therefore b_1 = b_5 = b_9 = b_{13} = \dots = \frac{1}{4}$$

$$\therefore b_2 = b_6 = b_{10} = b_{14} = \dots = 0$$

$\rightarrow (1.5 \text{ Marks})$



Sol(2)÷ Given signal, $x(t) = e^{-2|t-1|}$

$$\therefore X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} \cdot dt$$

$$= \int_{-\infty}^{\infty} e^{-2|t-1|} \cdot e^{-j\omega t} \cdot dt$$

$$= \int_{-\infty}^1 e^{2(t-1)} \cdot e^{-j\omega t} \cdot dt + \int_1^{\infty} e^{-2(t-1)} \cdot e^{-j\omega t} \cdot dt$$

$$= \int_{-\infty}^1 e^{-2} \cdot e^{(2-j\omega)t} \cdot dt + \int_1^{\infty} e^2 \cdot e^{(-2-j\omega)t} \cdot dt$$

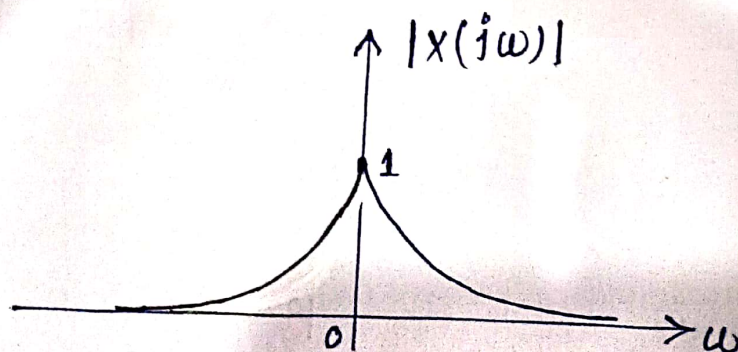
$$= e^{-2} \left\{ \frac{e^{(2-j\omega)t}}{(2-j\omega)} \right\}_{-\infty}^1 + e^2 \left\{ \frac{e^{(-2-j\omega)t}}{(-2-j\omega)} \right\}_1^{\infty}$$

$$= \frac{e^{-2}}{(2-j\omega)} \left\{ e^{(2-j\omega)} - 0 \right\} + \frac{e^2}{(-2-j\omega)} \left\{ 0 - e^{(-2-j\omega)} \right\}$$

$$= \frac{e^{-j\omega}}{(2-j\omega)} + \frac{e^{-j\omega}}{(2+j\omega)}$$

$$= \left(\frac{4e^{-j\omega}}{4+\omega^2} \right)$$

→ (3 Marks)



→ (2 Marks)